Discretionary Note

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IF YOU USE THIS FILE TO CHEAT, YOU ARE NOT ONLY STUPID BUT YOU ARE CHEATING YOURSELF OUT OF THE ABILITY TO FALL IN LOVE WITH MATH. Furthermore, I am not smarter than you and my solutions did not always get a perfect score.

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STATS 242 HW 6

March 5, 2025

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1.

We first compute the first moment μ_1 of $X \sim \text{Geom}(p)$, which is given by $\mu_1 = \mathbb{E}[X] = \frac{1}{p}$. We now solve for the MoM estimator $\hat{p}$ where the observed sample moment $\hat{\mu_1} = \frac{1}{n}(X_1 + \cdots + X_n) = \bar{X}$ is equal to theoretical moment μ_1 with the estimated $\hat{p}$ parameter:

$$\hat{\mu_1} = \bar{X} = \frac{1}{\hat{p}}$$
$$\hat{p} = \frac{1}{\bar{X}}$$

Thus, $\hat{p} = \frac{1}{\bar{X}}$ is the MoM estimator $\hat{p}$ for p. We now compute the MLE of p. We first start by finding the log-likelihood function $\ell_n(p)$:

$$\ell_n(p) = \log[\text{lik}(p)] = \Sigma_{i=1}^n \log\ f(X_i|p) = \Sigma_{i=1}^n \log(p(1-p)^{x_i - 1}) = \Sigma_{i=1}^n \log(p) + (x_i - 1)\log(1-p)$$
$$= n\log(p) + \Sigma_{i=1}^n (x_i - 1)\log(1-p)$$

We now solve for the MLE $\hat{p}$ by solving $\ell_n'(\hat{p}) = 0$:

$$\ell_n'(\hat{p}) = 0$$
$$(n\log(\hat{p}) + \Sigma_{i=1}^n (x_i - 1)\log(1-\hat{p}))' = 0$$
$$\frac{n}{\hat{p}} - \Sigma_{i=1}^n \frac{(x_i - 1)}{1-\hat{p}} = 0$$
$$\frac{n}{\hat{p}} - \frac{1}{1-\hat{p}}(n\bar{X} - n) = 0$$
$$\frac{n(1-\hat{p}) - \hat{p}(n\bar{X} - n)}{\hat{p}(1-\hat{p})} = 0$$
$$n(1-\hat{p}) - \hat{p}(n\bar{X} - n) = 0$$
$$n - n\hat{p} - \hat{p}n\bar{X} + n\hat{p} = 0$$
$$n = n\hat{p}\bar{X}$$
$$1 = \hat{p}\bar{X}$$
$$\hat{p} = \frac{1}{\bar{X}}$$

Thus the MLE of p is $\hat{p} = \frac{1}{\bar{X}}$. We now find the sampling distribution of the MLE of p, under large n. We begin by computing the Fisher Information $I(p)$:

$$I(p) = -\mathbb{E}_p[\frac{\partial^2}{\partial p^2}\log\ f(X|p)] = -\mathbb{E}_p[\frac{\partial^2}{\partial p^2}(\log(p) + (x-1)\log(1-p))]$$
$$= -\mathbb{E}_p[\frac{\partial}{\partial p}(\frac{1}{p} + (1-x)\frac{1}{1-p})] = -\mathbb{E}_p[-\frac{1}{p^2} + \frac{(1-x)}{(1-p)^2}] = -(-\frac{1}{p^2} + \frac{1}{(1-p)^2} - \frac{\mathbb{E}_p[x]}{(1-p)^2})$$
$$= \frac{1}{p^2} - \frac{1}{(1-p)^2} + \frac{1}{p(1-p)^2} = \frac{(1-p)^2 - p^2 + p}{p^2(1-p)^2} = \frac{1-p}{p^2(1-p)^2} = \frac{1}{p^2(1-p)}$$

So this means that for the MLE $\hat{p}$, $\sqrt{n}(\hat{p}-p) \to \mathcal{N}(0, \frac{1}{I(p)})$ or $\sqrt{n}(\hat{p}-p) \to \mathcal{N}(0, p^2(1-p))$ meaning $\hat{p} \to \mathcal{N}(p, \frac{p^2(1-p)}{n})$ as $n \to \infty$.

2.

We first compute the MoM estimate of p. To do so, we compute the first moment μ_1 of $X \sim \text{NegBinom}(r, p)$ which is given by $\mu_1 = \mathbb{E}[X] = \frac{pr}{1-p}$. We now solve for the MoM estimator $\hat{p}$ where the observed sample moment $\hat{\mu_1} = \frac{1}{n}(X_1 + \cdots + X_n) = \bar{X}$ is equal to theoretical moment μ_1 with the estimated $\hat{p}$ parameter:

$$\hat{\mu_1} = \bar{X} = \frac{\hat{p}r}{1-\hat{p}}$$
$$\bar{X}(1-\hat{p}) = \hat{p}r$$
$$\bar{X} = \hat{p}\bar{X} + \hat{p}r$$
$$\bar{X} = \hat{p}(\bar{X} + r)$$
$$\hat{p} = \frac{\bar{X}}{\bar{X}+r}$$

Thus the MoM estimator of p is given by $\hat{p} = \frac{\bar{X}}{\bar{X}+r}$. We now compute the MLE of p. We first start by finding the log-likelihood function $\ell_n(p)$:

$$\ell_n(p) = \log[\text{lik}(p)] = \Sigma_{i=1}^n \log\ f(x_i|p) = \Sigma_{i=1}^n \log(\binom{x+r-1}{x_i}(1-p)^r p^x)$$
$$= \Sigma_{i=1}^n \log(\binom{x_i+r-1}{x_i}) + r\log(1-p) + x_i\log(p) = nr\log(1-p) + \Sigma_{i=1}^n \log(\binom{x_i+r-1}{x_i}) + x_i\log(p)$$

We now solve for the MLE $\hat{p}$ by solving $\ell_n'(\hat{p}) = 0$:

$$\ell_n'(\hat{p}) = 0$$
$$(nr\log(1-\hat{p}) + \Sigma_{i=1}^n \log(\binom{x_i+r-1}{x_i}) + x_i\log(\hat{p}))' = 0$$
$$\frac{-nr}{1-\hat{p}} + \Sigma_{i=1}^n \frac{x_i}{\hat{p}} = 0$$
$$-\frac{nr\hat{p}}{\hat{p}(1-\hat{p})} + \frac{n\bar{X}(1-\hat{p})}{\hat{p}(1-\hat{p})} = 0$$
$$\frac{n\bar{X}(1-\hat{p}) - nr\hat{p}}{\hat{p}(1-\hat{p})} = 0$$
$$n\bar{X}(1-\hat{p}) - nr\hat{p} = 0$$
$$\bar{X}(1-\hat{p}) - r\hat{p} = 0$$
$$\bar{X} - \hat{p}\bar{X} - \hat{p}r = 0$$
$$\bar{X} = \hat{p}(\bar{X} + r)$$
$$\hat{p} = \frac{\bar{X}}{\bar{X}+r}$$

Thus the MLE of p is given by $\hat{p} = \frac{\bar{X}}{\bar{X}+r}$. We now find the sampling distribution of the MLE of p, under large n. We begin by computing the Fisher Information $I(p)$:

$$I(p) = \text{Var}_p[\frac{\partial}{\partial p}\log\ f(x|p)] = \text{Var}_p[\frac{\partial}{\partial p}[\log(\binom{x+r-1}{x}) + r\log(1-p) + x\log(p)]]$$
$$= \text{Var}_p[\frac{-r}{1-p} + \frac{x}{p}] = \frac{1}{p^2}\text{Var}_p[x] = \frac{r}{p(1-p)^2}$$

So this means that for the MLE $\hat{p}$, $\sqrt{n}(\hat{p}-p) \to \mathcal{N}(0, \frac{1}{I(p)})$ or $\sqrt{n}(\hat{p}-p) \to \mathcal{N}(0, \frac{p(1-p)^2}{r})$ so $\hat{p} \to \mathcal{N}(p, \frac{p(1-p)^2}{nr})$.

3.

(a) The PDF of Pareto(α, x_m) is given by $f(x|\alpha, x_m) = \frac{\alpha x_m^\alpha}{x^{\alpha+1}}$. Thus, the PDF of Pareto$(\theta, 1)$ is given by:

$$f(x|\theta, 1) = \frac{\theta \cdot 1^\theta}{x^{\theta+1}} = \frac{\theta}{x^{\theta+1}}$$

Expressed differently, we can set functions $T(x) = \log(\frac{1}{x}), A(\theta) = \log(\frac{1}{\theta}), h(x) = \frac{1}{x}$ so the PDF

$$f(x|\theta) = e^{\theta T(x) - A(\theta)}h(x) = \frac{e^{\theta T(x)}}{xe^{A(\theta)}} = \frac{(e^{\log(\frac{1}{x})})^\theta}{xe^{\log\frac{1}{\theta}}} = \frac{1}{x^\theta \cdot x \cdot \frac{1}{\theta}} = \frac{\theta}{x^{\theta+1}}$$

is identical to the aforementioned derived Pareto$(\theta, 1)$ PDF.

(b) We differentiate both sides of the identify $1 = \int_{\mathcal{X}} f(x|\theta)dx$ to compute $\mathbb{E}_\theta[T(X)]$:

$$\int_{\mathcal{X}} f(x|\theta)dx = 1$$
$$\frac{d}{d\theta}\int_{\mathcal{X}} f(x|\theta)dx = \int_{\mathcal{X}} \frac{d}{d\theta}f(x|\theta)dx = \frac{d}{d\theta}(1) = 0$$
$$\int_{\mathcal{X}} \frac{d}{d\theta}[e^{\theta T(x) - A(\theta)}h(x)]dx = 0$$
$$\int_{\mathcal{X}} [(e^{\theta T(x) - A(\theta)})'h(x) + h'(x)e^{\theta T(x) - A(\theta)}]dx = \int_{\mathcal{X}} [(e^{\theta T(x) - A(\theta)})(T(x) - A'(\theta))h(x) + 0]dx = 0$$
$$\int_{X} [e^{\theta T(x) - A(\theta)}h(x)](T(x) - A'(\theta))dx = 0$$
$$\int_{\mathcal{X}} f(x|\theta)(T(x) - A'(\theta))dx = \mathbb{E}_\theta[T(x) - A'(\theta)] = 0$$
$$\mathbb{E}_\theta[T(x)] = \mathbb{E}_\theta[A'(\theta)] = A'(\theta)$$

So we have $\mathbb{E}_\theta[T(x)] = A'(\theta)$. We now verify this is the case for our Pareto$(\theta, 1)$ model with its defined functions $T(x) = \log(\frac{1}{x}), A(\theta) = \log(\frac{1}{\theta}), h(x) = \frac{1}{x}$ in part (a). We first compute $\mathbb{E}_\theta[T(x)]$:

$$\mathbb{E}_\theta[T(x)] = \mathbb{E}_\theta[\log(\frac{1}{x})] = \int_{-\infty}^{\infty} f(x|\theta)\log(\frac{1}{x})dx = \int_1^{\infty} \frac{\theta}{x^{\theta+1}}\log(\frac{1}{x})dx$$

We can use a u-substitution of $u = \frac{1}{x}$ so $-u^{\theta-1}du = -\frac{x}{x^\theta} \cdot -\frac{1}{x^2}dx = \frac{1}{x^{\theta+1}}dx$. Thus, this integral can be re-written as:

$$\int_1^0 \log(u) \cdot \theta \cdot (-u^{\theta-1})du = \theta \int_0^1 \log(u) \cdot u^{\theta-1}du$$

Using integration by parts, we can set $a = \log(u), da = \frac{1}{u}du, db = u^{\theta-1}du, b = \frac{1}{\theta}u^\theta$. As such, we get:

$$\theta \int_0^1 \log(u) \cdot u^{\theta-1}du = \theta[ab\Big|_{u=0}^{u=1} - \int_0^1 bda]$$

$$= \theta[\frac{\log(u) \cdot u^\theta}{\theta}]\Big|_0^1 - \theta \int_0^1 \frac{1}{\theta}u^{\theta-1}du = 0 - \int_0^1 u^{\theta-1}du = -\frac{1}{\theta}u^\theta\Big|_0^1 = -\frac{1}{\theta} + 0 = -\frac{1}{\theta}$$

So $\mathbb{E}_\theta[T(x)] = -\frac{1}{\theta}$. We now compute $A'(\theta)$:

$$A'(\theta) = \frac{1}{\frac{1}{\theta}} \cdot \frac{-1}{\theta^2} = \frac{-\theta}{\theta^2} = -\frac{1}{\theta}$$

So because $\mathbb{E}_\theta[T(x)] = -\frac{1}{\theta} = A'(\theta)$, we have verified for our Pareto model in part (a) that $\mathbb{E}_\theta[T(x)] = A'(\theta)$.

(c) We first find the log-likelihood function $\ell_n(\theta)$:

$$\ell_n(\theta) = \log[\text{lik}(\theta)] = \Sigma_{i=1}^n \log f(x_i|\theta) = \Sigma_{i=1}^n \log[e^{\theta T(x_i) - A(\theta)}h(x_i)] =$$
$$\Sigma_{i=1}^n \log[e^{\theta T(x_i)}] - \log[e^{A(\theta)}] + \log[h(x_i)] = -nA(\theta) + \Sigma_{i=1}^n \theta T(x_i) + \log[h(x_i)]$$

To get the MLE of θ we solve for the solution to $0 = \ell_n'(\theta)$:

$$\ell_n'(\theta) = 0$$
$$(-nA(\theta) + \Sigma_{i=1}^n \theta T(x_i) + \log[h(x_i)])' = 0$$
$$-nA'(\theta) + \Sigma_{i=1}^n T(x_i) = 0$$
$$nA'(\theta) = \Sigma_{i=1}^n T(x_i)$$
$$A'(\theta) = \frac{1}{n}\Sigma_{i=1}^n T(x_i)$$
$$\mathbb{E}_\theta[T(x)] = A'(\theta) = \frac{1}{n}\Sigma_{i=1}^n T(x_i)$$

So therefore the MLE of θ is the unique solution to $\mathbb{E}_\theta[T(x)] = \frac{1}{n}\Sigma_{i=1}^n T(x_i)$. Because the generalized MoM estimator of θ is also the solution to $\mathbb{E}_\theta[T(x)] = \frac{1}{n}\Sigma_{i=1}^n T(x_i)$, and we are guaranteed this solution is unique, the generalized MoM estimator of θ is equivalent to the MLE of θ.

An example of this occuring would be the generalized MoM estimator based on $T(x) = \log(x)$ for the Pareto$(\theta, 1)$ model coinciding with the MLE.

4.

(a) By the CLT, $\sqrt{n}(\hat{p}-p) \to \mathcal{N}(0, p(1-p))$. Informally, this gives us that $\text{Var}[\sqrt{n}(\hat{p}-p)] \approx \hat{p}(1-\hat{p}) \implies n\text{Var}(\hat{p}-p) \approx \hat{p}(1-\hat{p}) \implies \text{Var}(p) \approx \frac{\hat{p}(1-\hat{p})}{n}$. Using this approximation for the variance of p, we can compute a 95% confidence interval for p as $\hat{p} \pm z^{\frac{0.05}{2}}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ where $z^{\frac{0.05}{2}}$ is the upper-$\frac{0.05}{2}$ point of the standard normal distribution.

(b) We solve $\sqrt{n}(\hat{p}-p) = \pm\sqrt{p(1-p)}z^{(\alpha/2)}$ below to create a confidence interval for p:

$$\begin{aligned}
\sqrt{n}(\hat{p}-p) &= \pm\sqrt{p(1-p)}z^{(\alpha/2)} \\
n(\hat{p}-p)^2 &= p(1-p)(z^{(\alpha/2)})^2 \\
n\hat{p}^2 - 2np\hat{p} + np^2 &= (z^{(\alpha/2)})^2 p - (z^{(\alpha/2)})^2 p^2 \\
(n + (z^{(\alpha/2)})^2)p^2 - (2n\hat{p} + (z^{(\alpha/2)})^2)p + n\hat{p}^2 &= 0
\end{aligned}$$

Using the quadratic formula, we construct the following confidence interval for p:

$$p = \frac{(2n\hat{p} + (z^{(\alpha/2)})^2) \pm \sqrt{(2n\hat{p} + (z^{(\alpha/2)})^2)^2 - 4n\hat{p}^2(n + (z^{(\alpha/2)})^2)}}{2(n + (z^{(\alpha/2)})^2)}$$

(c) For our confidence interval in part (a), we report the simulated coverage probabilities below:

	$n = 10$	$n = 40$	$n = 100$
$p = 0.1$	0.65061	0.91435	0.93267
$p = 0.3$	0.83926	0.92955	0.94984
$p = 0.5$	0.8922	0.91919	0.94205

We similarly report the simulated coverage probabilities for our confidence interval in part (b):

	$n = 10$	$n = 40$	$n = 100$
$p = 0.1$	0.93053	0.94406	0.9375
$p = 0.3$	0.92463	0.94432	0.93659
$p = 0.5$	0.97861	0.961	0.94316

We can clearly see from comparing these two tables that our interval construction in part (b) leads to generally much higher simulated coverage probabilities.

```python
        # %%
import numpy as np
from scipy.stats import norm

```

```python
z_alpha = np.abs(norm.ppf(0.05/2))
z_alpha_squared = z_alpha ** 2

NUM_SIMULATIONS = 100 * 1000
SAMPLE_SIZES = [10, 40, 100]
TRUE_PARAMS = [0.1, 0.3, 0.5]

def generate_data(n,p):
    return np.random.binomial(n=n, p=p) # number of successes

def part_a_interval(num_successes, n):
    p_hat = num_successes / n
    delta =  z_alpha * np.sqrt(p_hat * (1 - p_hat) / n)
    return p_hat - delta, p_hat + delta

def part_b_interval(num_successes, n):
    p_hat = num_successes / n
    denominator = 2 * (n + z_alpha_squared)
    center = (2 * n * p_hat + z_alpha_squared) / denominator
    delta = np.sqrt((2 * n * p_hat + z_alpha_squared) ** 2 - 4 * n * (p_hat ** 2) * (n + z_alpha_squared))
    delta /= denominator
    return center - delta, center + delta

# %%


for n in SAMPLE_SIZES:
    for p in TRUE_PARAMS:
        NUM_COVERAGE_A = 0
        for _ in range(NUM_SIMULATIONS):
            n_successes = generate_data(n, p)
            lower, upper = part_a_interval(n_successes, n)

            if p >= lower and p <= upper: NUM_COVERAGE_A += 1

        print(f"[PART (A)] For sample size n={n} and p={p}, simulated coverage probability: {NUM_COVERAGE_A / NUM_SIMULATIONS}")



for n in SAMPLE_SIZES:
    for p in TRUE_PARAMS:
        NUM_COVERAGE_B = 0
        for _ in range(NUM_SIMULATIONS):
            n_successes = generate_data(n, p)
            lower, upper = part_b_interval(n_successes, n)
```

```python

            if p >= lower and p <= upper: NUM_COVERAGE_B += 1

        print(f"[PART (B)] For sample size n={n} and p={p}, simulated coverage probability: {NUM_COVERAGE_B / NUM_SIMULATIONS}")


# %%
```